

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

M.Tech. (CAD/CAM)

Sem - I University Examination, 2009-10

ME – 703 Advance Dynamics of Machines

Date: 28.12.2009, Monday

Time: 01:00 p.m. to 04:00 p.m.

Maximum Marks: 70

Instructions:

1. Attempt all questions.
2. Figures to the right indicate full marks.
3. Assume suitable additional data if required stating clearly the assumptions made in your answer book.
4. Illustrate your answer with neat sketches wherever required.

SECTION- I

- Q1 (a) Explain the Kutzbach's criterion for spatial mechanism 05
- (b) Explain how will you consider effect of sliding friction in static force Analysis 05

- Q2 (a) Explain the analytical method of balancing of V-Engine. What is the value of resultant primary force for 60° V-Engine in terms mass, angular speed, crank radius and angle θ . 08
- (b) Investigate the state of primary and secondary balancing of a four stroke cycle four cylinder In-line engine with firing order I-II-III-IV. What will be the change in this state when firing order is altered to I-II-IV-III? 05

OR

- Q2 (a) The firing order of a 2-stroke cycle six cylinder diesel engine is I-V-III-VI-II-IV. The adjacent cylinder centerlines are 75 cm apart. Each cylinder has a connecting rod 120cm long and a stroke 45cm. The mass of the reciprocating parts in each cylinder is 200 kg. Determine the magnitudes of the primary and secondary unbalanced forces and moments when the engine runs at a constant speed of 250 rpm. 08
- (b) Investigate the state of unbalance of two stroke radial engine with six cylinders. 05

- Q3 (a) Explain the Analytical Method for Dynamic force Analysis. 04
- (b) Determine static equilibrium torque C_2 on link O_2A (Fig 1.) by Analytical method where $O_2A = 15\text{cm}$ $AB = 45\text{cm}$ and $AC = 12\text{cm}$ angle $AO_2B = 45^\circ$. 08

OR

- Q3 (a) The dimensions of a four-link mechanism are, $AB=500\text{ mm}$, $BC= 660\text{ mm}$, $CD=560\text{ mm}$ and $AD= 1000\text{ mm}$. The link AB has an angular velocity of 10.5 rad/s counter-clockwise and angular retardation of 26 rad/s^2 at the instant when it makes an angle of 60° with AD, the fixed link. The mass of the links BC and CD is 4.2 Kg/m length. The link AB has a mass of 3.54 Kg , the center of which lies at 200mm from A and a mass moment of inertia of 88500 kg-mm^2 . Neglect the gravity and friction effects, determine the instantaneous value of the drive torque required to be applied on AB to overcome the inertia forces by analytical method. 12

P.T.O

SECTION - II

- Q4 (a)** Design and draw a slider crank mechanism in which displacement of slider S is given by following relation :

$$S_j = \frac{66825 + 630\theta_j - 7\theta_j^2}{8100}$$

Where θ_j = Crank angle of rotation in the range, $45^\circ \leq \theta \leq 135^\circ$. Use three point Chebyshev spacing.

- Q5 (a)** Fig.2 shows a quick return mechanism. Determine velocity and acceleration of link 4 (O_4A), link 5 (BC) and link 6 (Slider). Crank rotates at 200 r.p.m. $O_2A=140\text{mm}$, $O_4B=800\text{mm}$, $BC=250\text{mm}$, $\angle AO_2O_4 = 120^\circ$. Use vector method. 12

OR

- Q5 (a)** ABCD is a four bar mechanism, link length $AB = 50\text{ mm}$, $BC = 90\text{ mm}$, $CD = 110\text{ mm}$, fixed link $DA = 125\text{ mm}$ and $\angle BAD = 50^\circ$. If the link AB rotates at constant angular velocity of 12 rad/sec . Determine velocity and acceleration of other links by any **two** analytical methods and compare the results. 12

- Q6 (a)** How will you determine jump speed for an eccentric cam operating a flat faced follower? Prove the equation. 05

- (b)** An eccentric plate cam of 160 mm diameter and eccentricity of 40 mm provides motion to a spring loaded follower of mass 2 Kg whose axis is perpendicular to the axis of the cam and passes through its center. The spring has stiffness of 25 N/mm . It is found that at certain speed, the follower ceases to have contact with cam, when cam has moved through 110° from its lowest position. Determine this speed. The initial compression of the spring is 32 mm . Determine also the limiting angular speed of the cam to avoid cam jump. 08

OR

- Q6 (a)** Explain Polydyne cam, give design steps with advantages and limitations. 03

- (b)** Explain cam curvature and undercutting in roller follower. 04

- (c)** A radial cam with translating flat faced follower turns clockwise with the follower rising 20 mm with SHM in 160° of cam rotation. The base circle is 75 mm in diameter. Find the radial distance and angular follower location after the cam has rotated 120° from initial position. The cam rotates at 300 rpm . 06

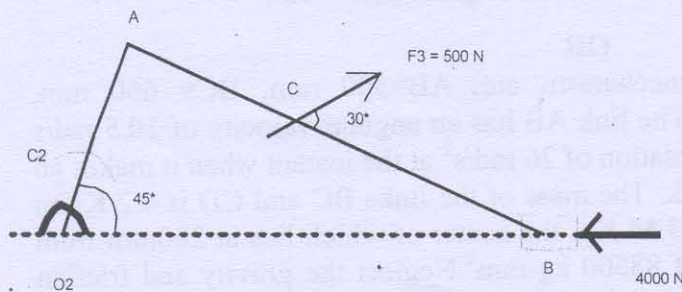


Fig.1

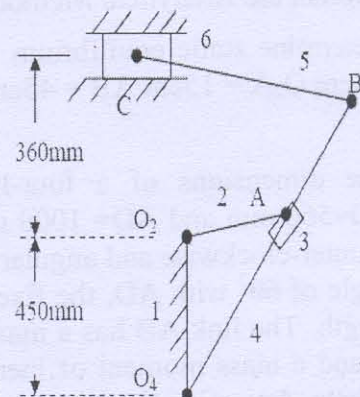


Fig.2

Figure is not to the scale.